

Example project-A product purchase triggers sending an email.

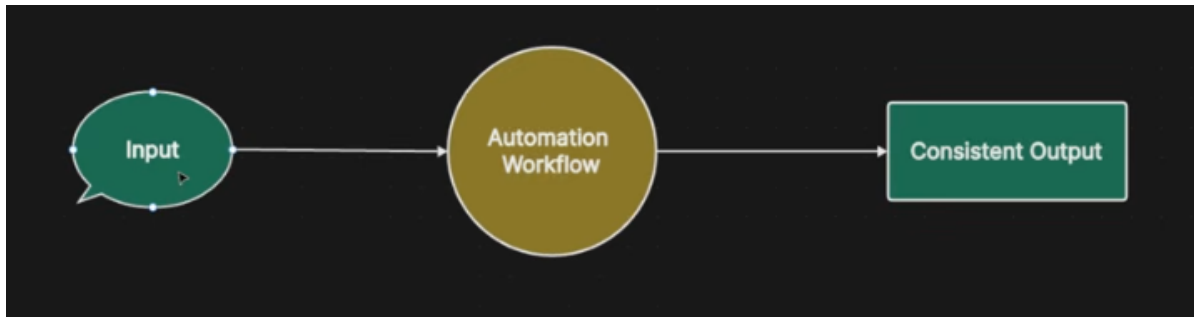
https://www.youtube.com/watch?v=ZblVOy_GPyQ&t=989s

What Are Agents and Agentic Systems?

- **Agentic systems** = environments built from:
 - **Agents:** Dynamic automations that *decide* what tools to use and what output to generate.
 - **Workflows:** Automations with *predefined* outputs.
- **Workflows Example:**
 - A product purchase triggers sending an email.
- **Agents Example:**
 - A customer support agent dynamically decides to:
 - Create a support ticket.
 - Pull order history.
 - Issue a refund.

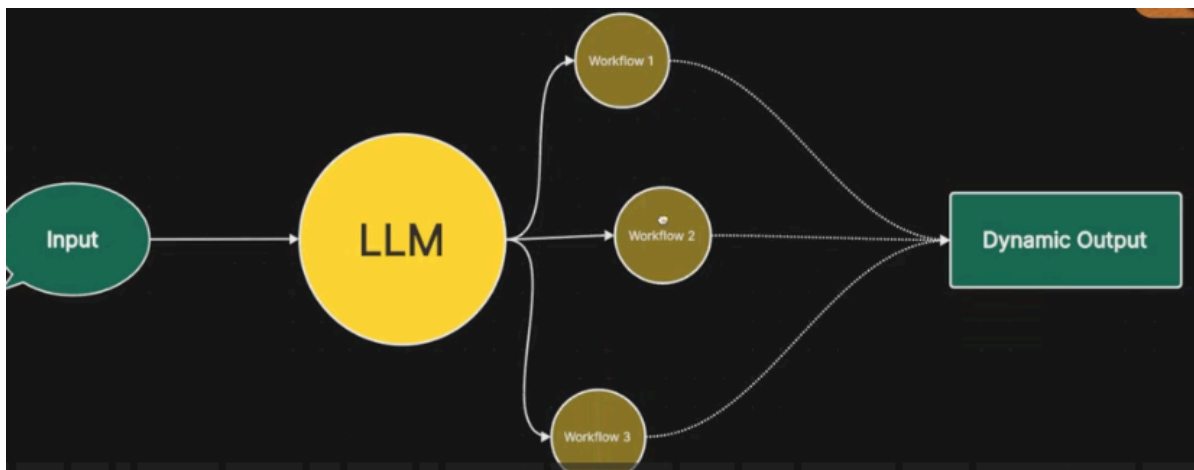
Workflows vs. Agents (Visual Difference)

- **Workflow:**
 - Input → Predefined steps → Consistent output destination.



- **Agent:**

- Input → Language model decides:
 - Which tool(s) to use.
 - What output to produce.
- Output is *dynamic*.



- Agents can *call other agents* with their own tools (nesting).

● N8N Overview

- **N8N:** No-code/low-code workflow automation platform.
- **Main sections:**
 - **Workflows:** Automations you build.
 - **Credentials:** API keys and access tokens.
 - **Executions:** Runs of your workflows.

Five Node Categories in N8N

1. Triggers:

- Start automations (e.g., on chat message, scheduled event, app event).

2. Action Nodes:

- Do something in an app/service (Google Sheets, Airtable, Gmail).

3. Utility Nodes:

- Transform data (IF statements, filters, conversions).

4. Code Nodes:

- Run code (JavaScript), HTTP requests, webhooks.

5. Advanced AI Agent Node:

- Makes workflows *autonomous*:
 - LLM chains.
 - Memory.
 - Retrieval-augmented generation.
 - Sentiment analysis.

Workflow Pipeline

Creating a Trigger

- Multiple trigger types possible.
- Example:
 - **On Chat Message Trigger:**
 - Public or private.
 - Provides a test URL and embedded chat options.

Adding the AI Agent Node

- **AI Agent Node** = Brain of your automation.
 - Needs a **Chat Model** to power it:
 - Example: OpenAI GPT-4.
 - Other options: Anthropic, AWS Bedrock, Grok, etc.
 - **Credential setup** (OpenAI example):
 - Get API key from platform.openai.com.
 - Paste into N8N credentials.
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Testing the AI Agent

- You can chat with it like ChatGPT.
 - *Without memory*, it cannot track context across messages.
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Adding Memory

- Use **Window Buffer Memory**:
 - Stores recent messages (context window length).
 - Session key ties memory to conversation.
 - Result: Agent can track sequences and context.
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Adding Tools to the AI Agent

- Tools = capabilities the agent can dynamically decide to use.
 - Example Tool:
 - **Airtable Search** (product catalog database).
 - **Tool Description**:
 - Important to clearly explain what the tool does.
 - **Operations**:
 - Search, create, update, delete records.
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Airtable Tool Setup

- **Credential:**
 - Create Airtable access token (scopes: data read/write, schema).
 - Link base (e.g., Home Inventory).
 - **Use Cases:**
 - Ask "What's out of stock?" → Agent searches Airtable.
 - Ask "What's close to reorder threshold?" → Dynamic retrieval.
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